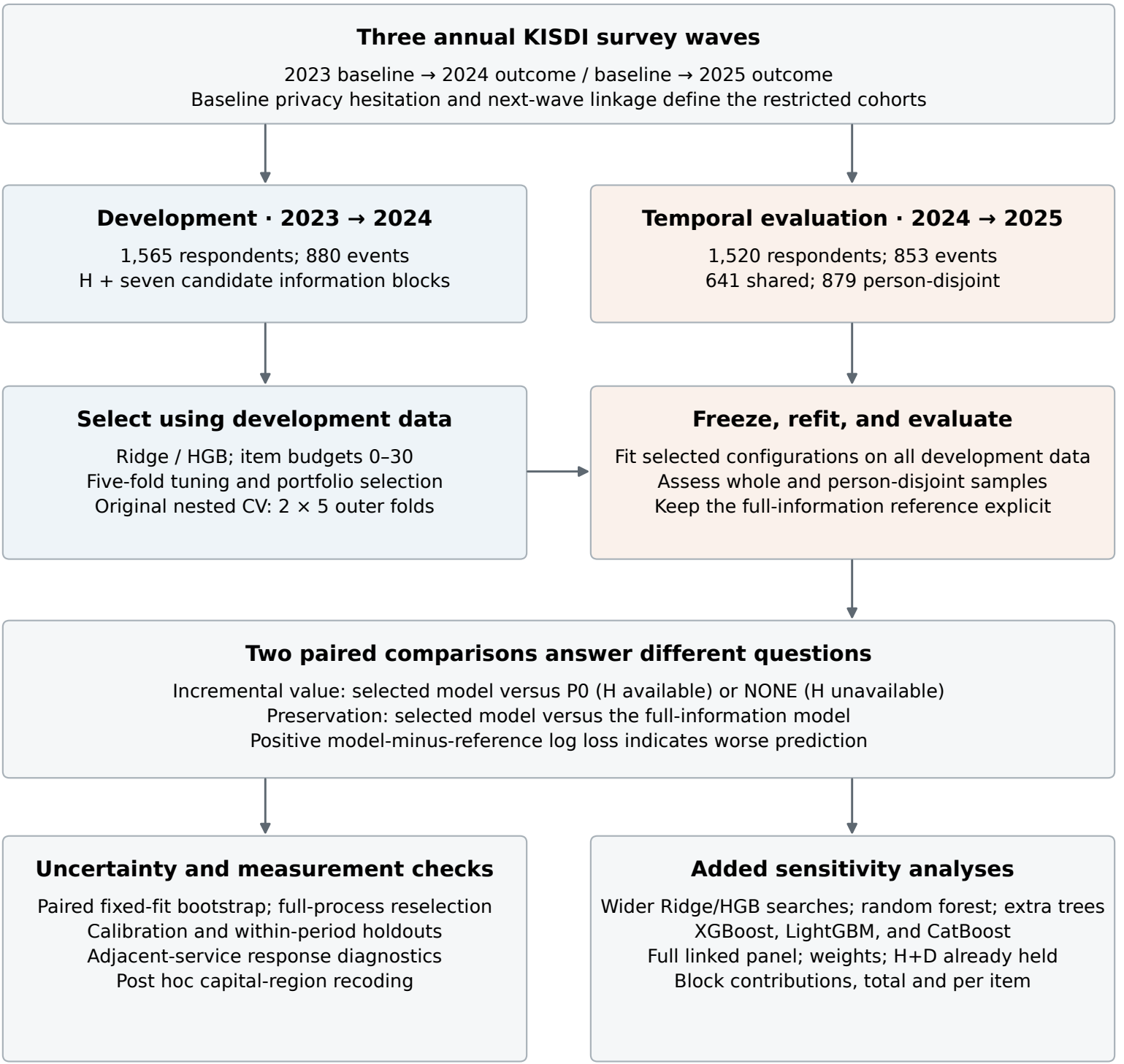
Three annual KISDI survey waves
2023 baseline → 2024 outcome / baseline → 2025 outcome
Baseline privacy hesitation and next-wave linkage define the restricted cohorts
Development · 2023 → 2024
1,565 respondents; 880 events
H + seven candidate information blocks
Temporal evaluation · 2024 → 2025
1,520 respondents; 853 events
641 shared; 879 person-disjoint
Select using development data
Ridge / HGB; item budgets 0–30
Five-fold tuning and portfolio selection
Original nested CV: 2 × 5 outer folds
Freeze, refit, and evaluate
Fit selected configurations on all development data
Assess whole and person-disjoint samples
Keep the full-information reference explicit
Two paired comparisons answer different questions
Incremental value: selected model versus P0 (H available) or NONE (H unavailable)
Preservation: selected model versus the full-information model
Positive model-minus-reference log loss indicates worse prediction
Uncertainty and measurement checks
Paired fixed-fit bootstrap; full-process reselection
Calibration and within-period holdouts
Adjacent-service response diagnostics
Post hoc capital-region recoding
Added sensitivity analyses
Wider Ridge/HGB searches; random forest; extra trees
XGBoost, LightGBM, and CatBoost
Full linked panel; weights; H+D already held
Block contributions, total and per item